

Transmission media

* Transmission media is a pathway that carries the information from sender to receiver.

we use different types of cables or waves to transmit data.

Data is transmitted normally through electrical or electromagnetic signal.

* Guided media

Unguided media

i) Also called as wired communication or bounded transmission media.

also called as wireless communication or unbounded transmission media

ii) Signal propagates through wire

Signal propagates through air.

iii) Used for point to point communication

Generally suited for radio broadcasting in all directions

iv) cost effective

Expensive

v) Discrete network topologies are formed

Continuous network topologies are formed.

vi) Signals are in form of voltage, current or photons

Signals are in form of electromagnetic waves

vii) Transmission media capacity can be increased

It is not possible to obtain additional capacity

viii) Best option for shorter distance

Suitable for longer distance

ix) Unable to pass through walls

Can pass through walls.

x] Ex- twisted pair cables, coaxial cable, & optical Fibre cables

Ex- microwave or radio wave links & infrared light.

2.1] Guided media

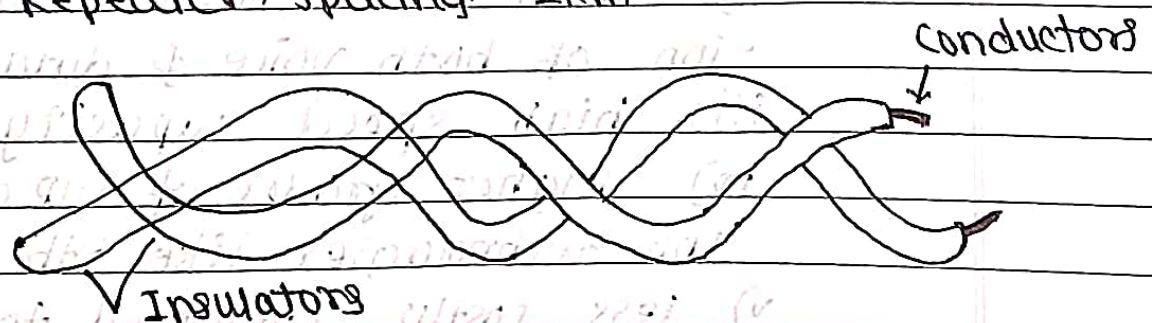
• signal directed & contained by physical limits of medium.

• Twisted pair } use metallic
coaxial cable } (copper) conductors

• Optical Fibre - accepts & transports signal in form of light.

Q* Twisted pair cable

- commonly used
- cheaper than others
- lightweight, can be installed easily
- have two conductors that are generally made up of copper & each conductor has insulation. These two conductors are twisted together, thus called as twisted pair cables.
- One conductor used to carry signal & other for ground reference only.
- Receiver uses difference of signals between these two conductors.
- The noise or crosstalk is greatly reduced due to twisting. In first twist, one conductor is near noise & other is far from source but in next twist the reverse happens & resultant noise is very less or no noise.
- Quality of signal in twisted pair cable depends on the no. of twists per unit length of cable.
- Frequency range - 0 to 3.5 KHz
- attenuation - 0.2 dB/km @ 1 KHz
- Typical delay - 50.2 μ s/km
- Repeater spacing - 2km

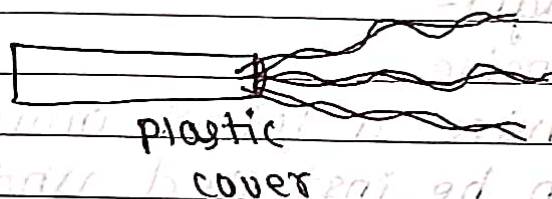


• Unshielded Twisted Pair Cables (UTP)

- Common type of telecommunication when compared with shielded twisted pair cable which consists of two conductors usually copper, each with its own colour each plastic insulator with its own plastic insulator.
- Copper wires twisted together without any other insulation or shielding.
- They are arranged in pairs so that we can add a new connection whenever required.
- When UTP are arranged in pairs, each pair is coded with different color as defined by 25 pair color code developed by AT & T corporation.
- The Electronic Industries Association divides UTP into 7 categories based upon cable quality where 1 is for highest quality & 7 is for lowest quality.
- Advantages:-
 - i) cost-effective, easy to install
 - ii) Used for short-distance transmission of both voice & data.
 - iii) high speed capacity
 - iv) Higher grades of UTP are used in LAN technologies like Ethernet
 - v) less costly compared to other types

• Disadvantages:-

- i) not secure
- ii) efficient only for distance upto 100 meters
- iii) limited bandwidth
- iv) ~~It~~ provides less protection from interference.



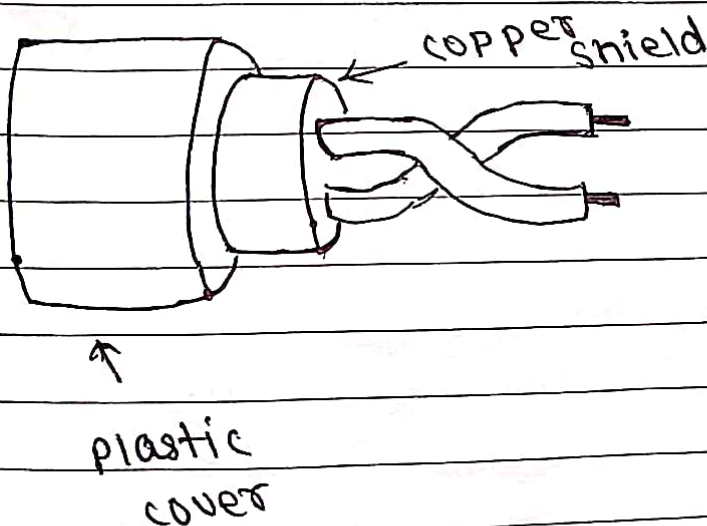
• Shielded Twisted Pair Cable

• These types of cables have extra insulation or protective covering over conductors in form of copper braid covering.

• This covering provides strength to overall structure of the cable.

• It also reduces noise & signal interference

• These are also color coded like UTP



• Advantages-

- used for long distance communication
- Protective shield prevents external electromagnetic noise penetration into the cable.
- higher bandwidth than UTP

• Disadvantages-

- expensive
- require a lot of maintenance
- can be installed underground only.

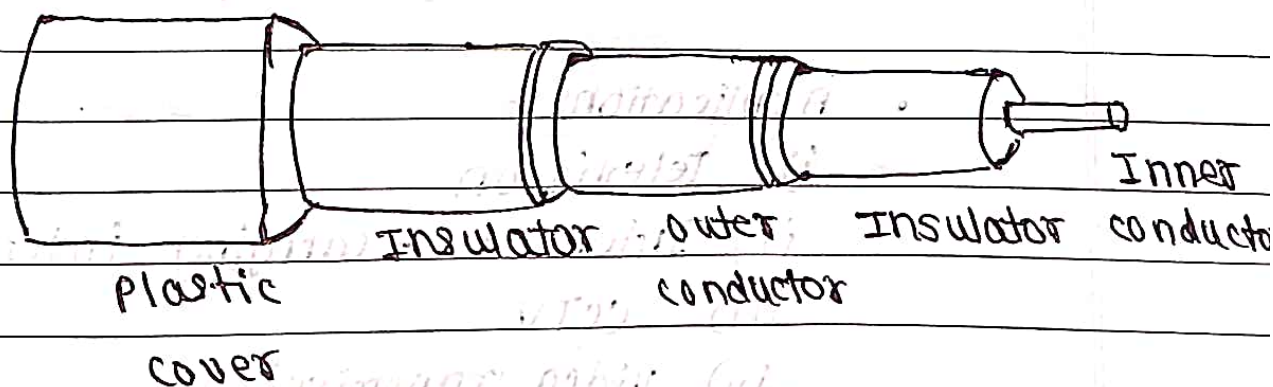
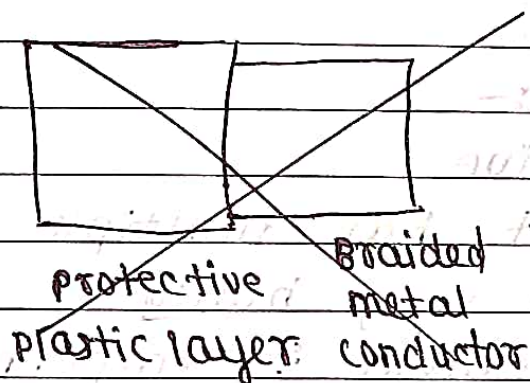
• Applications of Twisted Pair cables:

- i) telephone lines
- ii) DSL lines
- iii) LAN
- iv) can be used in both analog & digital transmission
- v) RJ-45 is it's common application.



2) COAXIAL CABLES:-

- Coaxial is called by this name becoz it contains two conductors that are parallel to each other.
- Copper in this is used as a centre conductor
- It is surrounded by PVC insulation, a sheath which is encased in an outer conductor of metal foil, braid or both.
- Outer metallic wrapping is used as a shield against noise & as the second conductor which completes the circuit
- Outermost part is plastic cover which protects the whole cable.



- two types :-

1) BaseBand

This is 50 ohm (Ω) coaxial cable which is used for digital transmission

2) BroadBand

This uses analog transmission on standard cable television cabling.

• Advantages :-

i) Bandwidth is high

ii) Used in ~~the~~ long distance telephone lines

iii) speed is high

iv) easy to install

v) less affected by noise or cross talk

vi) support multiple channels

• Disadvantages :-

i) expensive

ii) As it has multiple layers it is very bulky

iii) IF shield is imperfect, it can lead to grounded loop.

• Applications :-

i) Television

ii) Internet - carrying internet signals

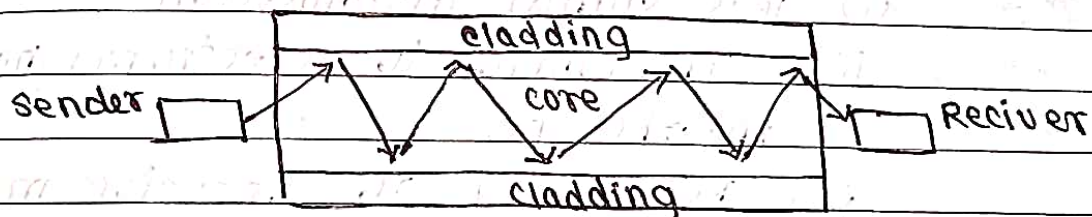
iii) CCTV

iv) video Transmission

v) HDTV

3] FIBER OPTIC CABLE:-

- made up of glass or plastic & transmits signals in form of light.
- A optical fibres uses reflection to guide light through a channel.
- A glass or plastic cover is surrounded by a cladding of less dense glass or plastic.
- The difference in density of two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it.



internal view of fiber optic cable

- on basis of no. of modes, it is classified into 2 types:-

a] single-mode fiber:-

Only one type of ray of light can propagate through the fiber.

b] Multi mode fiber:-

provides large no. of modes for light ray travelling through it.

- Multimode have two types:-

a) step index optical fiber:

Refractive index of core and cladding is constant.

b) Graded index optical fiber:

Core has a non-uniform refractive index that gradually decreases from the center towards the core cladding interface. The cladding has a uniform refractive index.

- Advantages :-

- i) Higher bandwidth
- ii) less signal attenuation
- iii) Immunity to electromagnetic interference
- iv) Resistance to corrosive materials
- v) light weight
- vi) Greater immunity to tapping

- Disadvantages :-

- i) Installation & maintenance
- ii) High cost
- iii) Undirectional light propagation

- Applications :-

- i) often used in backbone networks
- ii) some TV companies use combination of optical fibre & coaxial cable

2.2] UNGUIDED MEDIA :-

we can divided UNGUIDED MEDIA in three broad groups :-

1) Radio waves :-

- Electromagnetic waves ranging in Frequencies between 3 kHz to 1 GHz are normally called radio waves.

- Radio waves are omnidirectional. when an antenna transmits radio waves, they are propagated in all directions. This means that sending & receiving antennas don't have to be aligned.

- Sending antenna send waves that can be received by any receiving antenna.

The omnidirectional property has disadvantages too.

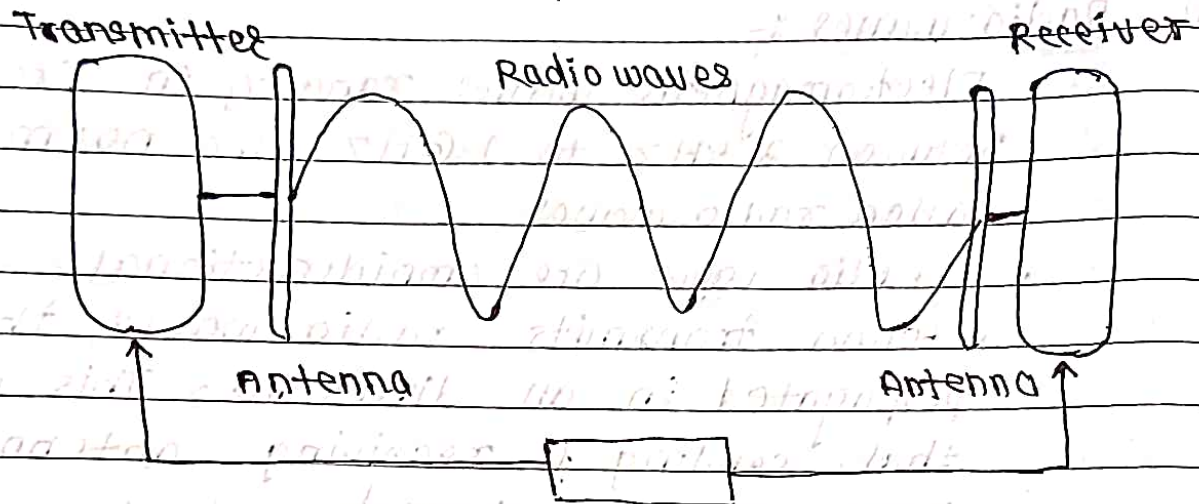
The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signal using the same Frequency or band.

- radio waves with low & medium frequency can penetrate walls.

It is advantageous becoz, an AM radio can receive signals inside a building. It is disadvantage becoz we can't isolate a communication to just inside or outside a building.

• Applications :-

- i) omnidirectional property is useful in multitasking
- ii) AM & FM radio, television, maritime radio, cordless phone.



2) Micro waves :-

- Electromagnetic waves having frequencies between 1 & 300 GHz are called micro waves. micro waves are unidirectional.
- When an antenna transmits microwaves, they can be narrowly focused. This means that the sending & receiving antennas need to be aligned.
- microwave propagation is line-of-sight.
- very high-frequency microwaves can't penetrate walls.
- The microwave band is relatively wide, almost 299 GHz.

• Applications :-

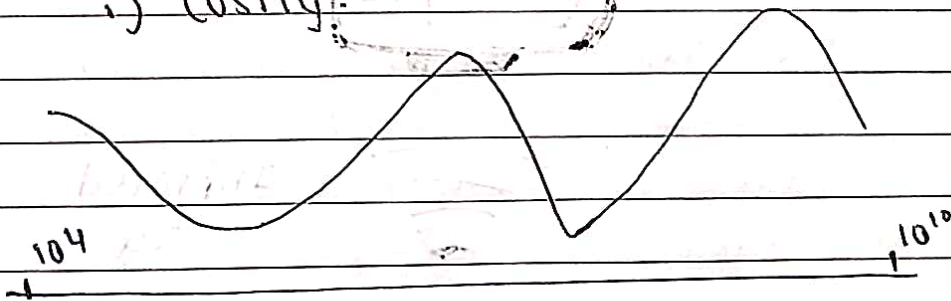
- i) microwaves, due to their unidirectional properties, are very useful in one-to-one communication is needed
- ii) used in cellular phones, satellite networks & wireless LANs.

• Advantages -

- i) used for long distance telephone communication
- ii) carries 1000's of voice channels at the same time

• Disadvantages -

- i) costly.



⑧ INFRARED WAVES :-

- Electromagnetic waves that have frequency range between 300 GHz to 400 GHz.
- They can't travel along long distances. These are used for short distance communication.
- These waves use line-of-sight of propagation.
- These waves can't penetrate through solid objects like walls etc.

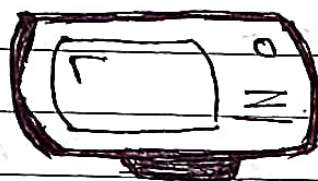
- we can't use infrared waves outside the building becoz the sun's rays contain infrared waves that can interfere with the communication.

• Applications -

i) Remote controls that are used for TV, DVD players and stereo system.

ii) communication between devices such as keyboards, mouse, PCs & printers.

iii) Used in short range communication



- microwaves communication -
 satellite microwave
~~Terrestrial~~ Terrestrial microwave

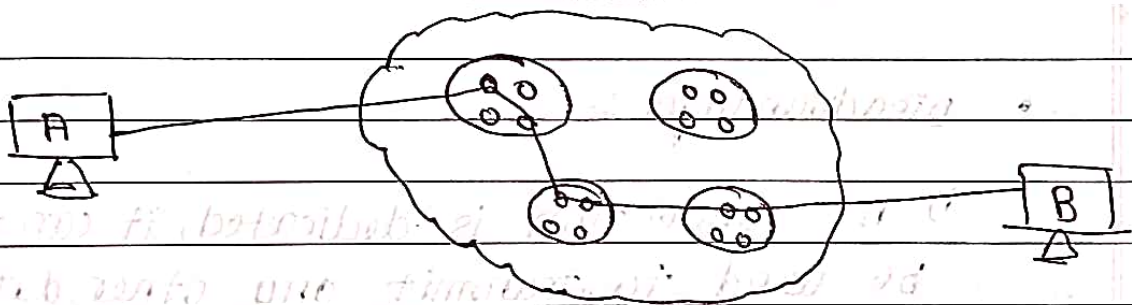
2.3] Switching Basics

Switched communication networks are those in which data transferred from source to destination is routed between various intermediate nodes.

switching is the technique by which nodes control or switch data to transmit it between specific points on a network.

~~It is~~ There are 3 common techniques-

1] Circuit Switching



- When two nodes communicate with each other over a dedicated communication path.
- There is a need of pre-specified route from which data will travel & no other data is permitted.
- In circuit switching a dedicated channel (or circuit) is set up for a single connection between the sender & recipient during the communication session. Once the connection is established between two parties, it will be available till end of the conversation. the channel is

reserved between the users till the connection is active.

- In telephone communication system, the normal voice call is the example of circuit switching. The telephone service provider maintain a unbroke link for each telephone call.

- Advantages :-

- i) Guaranteed data rate
- ii) Data is transmitted without any delay
- iii) Suitable for long continuous transmission.

- Disadvantages :-

- i) As connection is dedicated, it can't be used to transmit any other data even if channel is free.

- ii) Dedicated channels require more bandwidth

- iii) time consuming.

Q] message switching :-

1. This was developed as an alternative to circuit switching before packet switching.

- In msg switching, end-users communicate by sending & receiving msgs that included the entire data to be shared. Msgs are the smallest individual unit.
- Sender & receiver are not directly connected.

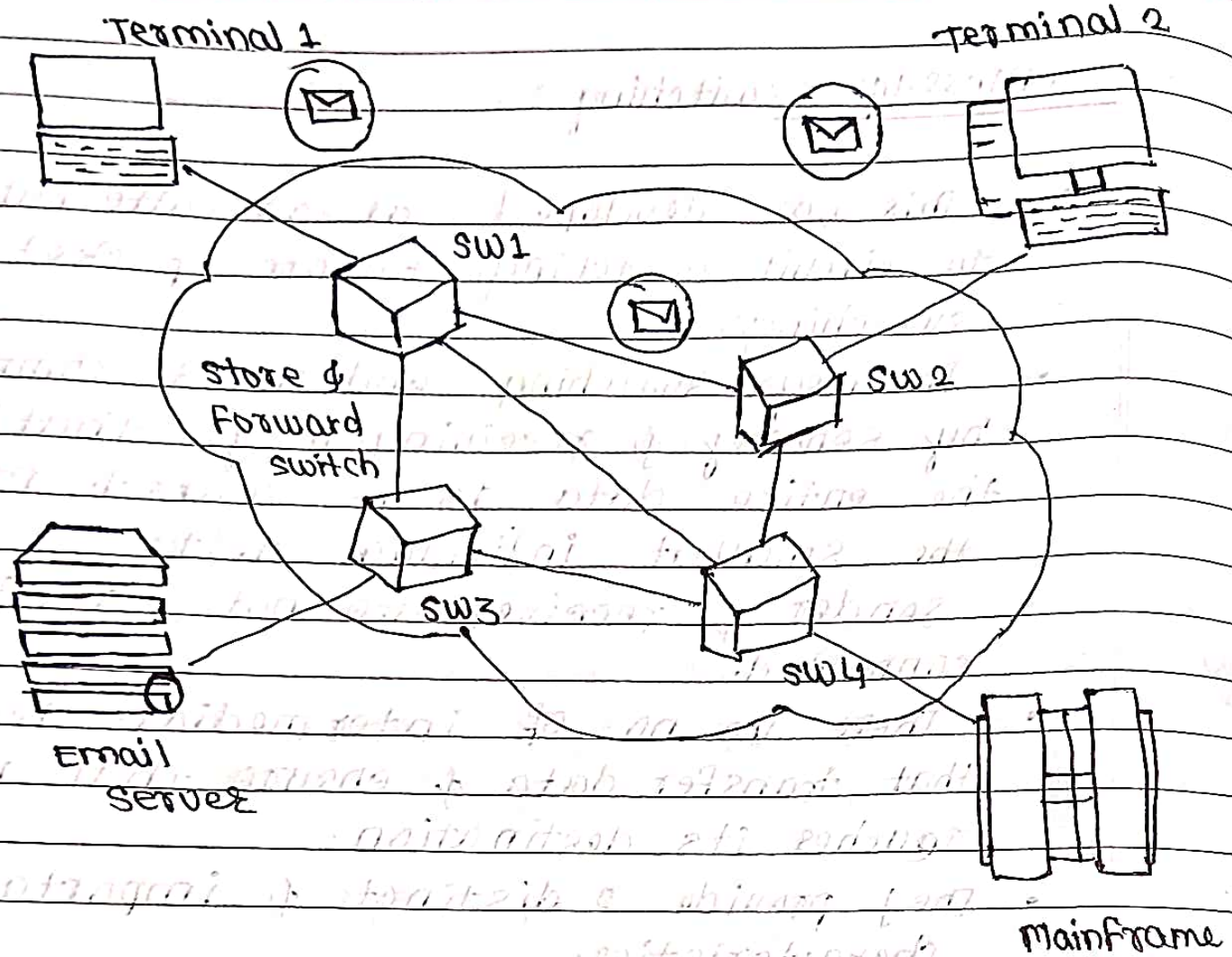
2. There are no. of intermediate nodes that transfer data & ensure that msg reaches its destination.

- They provide 2 distinct & important characteristics.

1) Store & Forward-

The intermediate nodes have responsibility of transferring the entire msg to the next node. Hence each node must have storage capacity. A store & Forward switch forwards a msg only if sufficient resources are available & next hop is accepting data.

2) msg delivery- This implies wrapping the entire info in single msg & transferring it from the source to the destination node.



• Advantages :-

- i) Helps in reducing the traffic congestion in the network.
- ii) data channels are shared by the network devices.
- iii) AS msgs are delivered via a store & forward method, it is possible to include priority in them.
- iv) It allows for infinite msg lengths.

• Disadvantages-

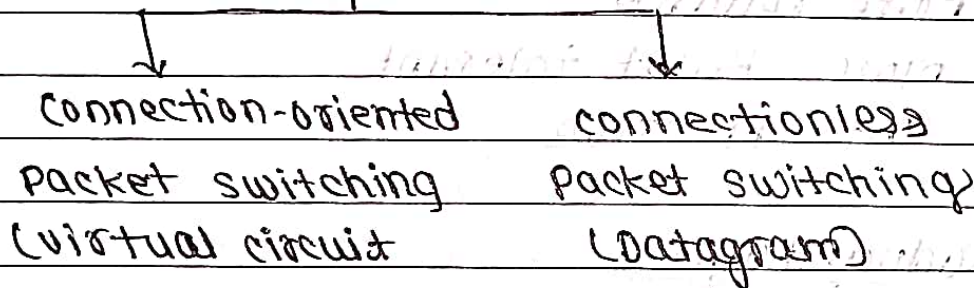
- i) can't store in real-time applications
- ii) requires large storing capacity
- iii) It is not dependable communication bcoz there is no direct relationship between sender & receiver.

• Applications-

- i) Telegraph msg switching centres
- ii) Delivery process in electronic mail systems
- iii) Delivery process in many major networks & systems.

Q

Packet switching



- Packet switching is a method of transferring the data to a network in form of packets.
- In order to transfer the file fast & efficiently manner over the network & minimize the transmission latency, the data is broken into small pieces of variable length, called packet.
- At the destination, all these small parts

(packets) have to be resembled, belonging to the same file.

- A packet composes of payload & various control information. No pre-setup or reservation of resources is needed.
- Packet switching uses store & forward technique while switching the packets.
- Packet switched networks were designed to overcome the weakness of circuit-switched networks.

• Advantages :-

(over circuit switching)

- i) more efficient in terms of bandwidth
- ii) minimal transmission latency
- iii) more reliable
- iv) more fault tolerant
- v) cost effective

• Disadvantages:-

i) Packet switching doesn't give packets in order.

ii) Since packets are unordered, we need to provide sequence no's for each packet.

iii) complexity is more.

iv) Transmission delay is more.

v) Beneficial for small msg but not for long.

• modes of Packet Switching :-

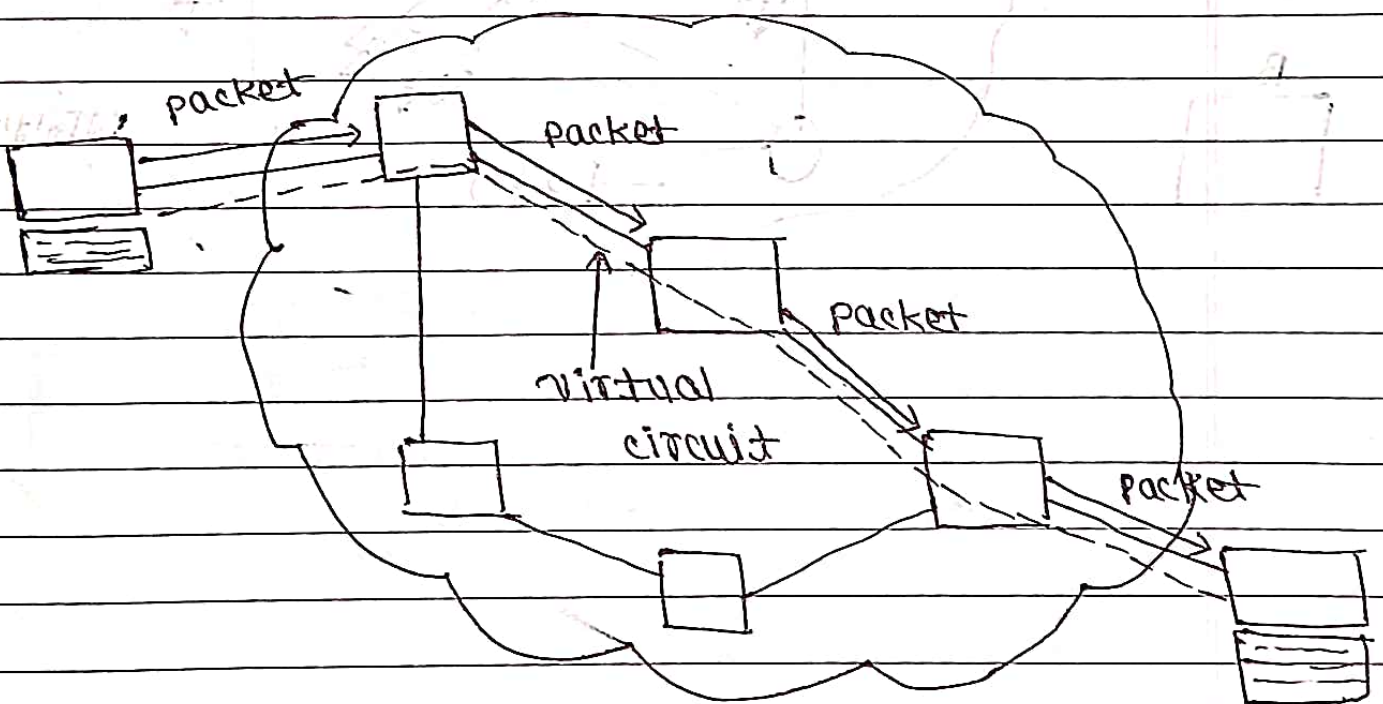
i) connection-oriented Packet Switching (virtual circuit) :-

i) Before starting the transmission, it establishes a logical path or virtual connection using signaling protocol, between sender & receiver & all packets belongs to this flow will follow this predefined route.

ii) virtual circuit ID is provided by switches/routers to uniquely identify this virtual connection.

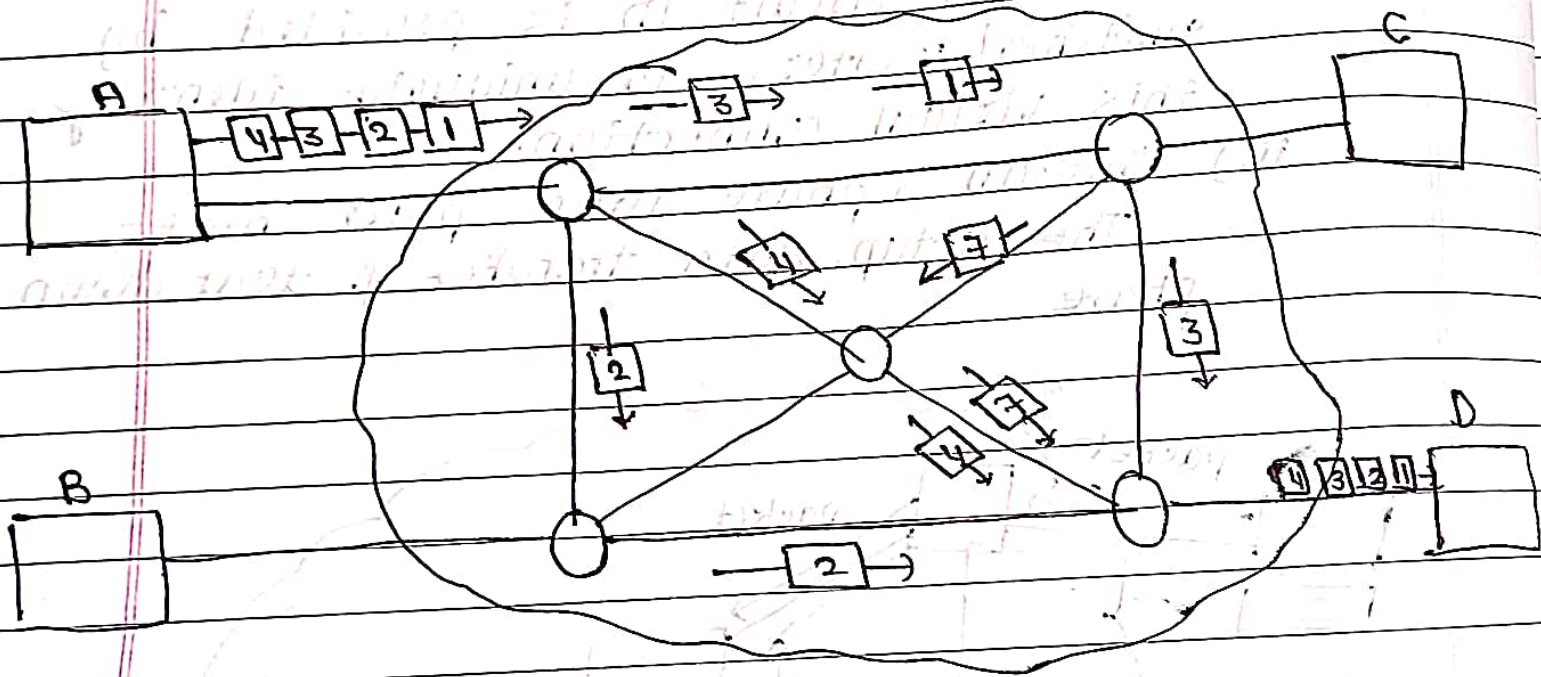
iii) Overall 3 phase take place here-

The setup, data transfer & tear down phase



a] Connectionless Packet Switching (Datagram):-

- Unlike connection-oriented packet switching, In connectionless packet switching each packet contains all necessary addressing information such as source address, destination address & port no's etc.
- In datagram packet switching; each packet is treated independently.
- It has no connection setup & teardown phase, like virtual circuits.



Circuit switching

Msg switching

Packet switching

1) There is physical connection b/w transmitter & receiver	no physical path — —	no physical path — —
2) All packets uses same path	Packet are stored & forward	Packet travels independently
3) need an end to end path before the data transmission	no need of an — —	no need of an — —
4) Reserves entire bandwidth in advance	Doesn't reserve bandwidth — —	Doesn't reserve bandwidth — —
5) waste of bandwidth is possible	no waste of bandwidth	no waste of bandwidth
6) It can't support store & forward transmission.	It support store — —	It support store & forward — —
7) not suitable for handling interactive traffic	Suitable For handling — —	Suitable For handling — —